WORKED SOLUTION

# STEP I 2011, Question 2

Pure • Integration by Substitution

**STEP I 2011, Question 2**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s1-q2

## The key idea

The integral $E$ has no elementary closed form, so we never try to evaluate it. Instead, every other integral is to be manipulated until $E$ itself appears. For the first two, this means splitting the fraction by algebraic division so that the troublesome $\frac{1}{1+x}$ is isolated. For (i) and (ii), it means choosing a substitution that turns the integrand back into $\frac{\mathrm{e}^u}{1+u}$ on $[0,1]$: look at the exponent and ask what would make it the new variable.

## Solution

### The first two integrals

Writing $x = (1+x) - 1$,

$$\frac{x\,\mathrm{e}^x}{1+x} = \mathrm{e}^x - \frac{\mathrm{e}^x}{1+x},$$

so

$$\int_0^1 \frac{x\,\mathrm{e}^x}{1+x}\,\mathrm{d}x = \int_0^1 \mathrm{e}^x\,\mathrm{d}x - E = \mathrm{e} - 1 - E,$$

as required.

For the second, divide $x^2$ by $1+x$: since $x^2 = (x-1)(1+x) + 1$,

$$\frac{x^2\,\mathrm{e}^x}{1+x} = x\,\mathrm{e}^x - \mathrm{e}^x + \frac{\mathrm{e}^x}{1+x}.$$

Integrating by parts with $u = x$, $\mathrm{d}v = \mathrm{e}^x\,\mathrm{d}x$,

$$\int_0^1 x\,\mathrm{e}^x\,\mathrm{d}x = [x\,\mathrm{e}^x]_0^1 - \int_0^1 \mathrm{e}^x\,\mathrm{d}x = \mathrm{e} - (\mathrm{e} - 1) = 1.$$

Hence

$$\int_0^1 \frac{x^2\,\mathrm{e}^x}{1+x}\,\mathrm{d}x = 1 - (\mathrm{e} - 1) + E = 2 - \mathrm{e} + E.$$

(Alternatively, write $\frac{x^2}{1+x} = x - \frac{x}{1+x}$ and use the first result directly: $1 - (\mathrm{e} - 1 - E)$ gives the same answer.)

**Part (i)**

The exponent is the awkward part, so we make it the new variable:

$$u = \frac{1-x}{1+x}.$$

Solving for $x$: $u(1+x) = 1-x$, so $x = \dfrac{1-u}{1+u}$. Then

$$1+x = \frac{2}{1+u}, \qquad \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{-(1+u)-(1-u)}{(1+u)^2} = -\frac{2}{(1+u)^2}.$$

When $x = 0$, $u = 1$; when $x = 1$, $u = 0$. Therefore

$$\begin{aligned}\int_0^1 \frac{\mathrm{e}^{\frac{1-x}{1+x}}}{1+x}\,\mathrm{d}x &= \int_1^0 \mathrm{e}^u \cdot \frac{1+u}{2} \cdot \left(-\frac{2}{(1+u)^2}\right)\mathrm{d}u \\ &= \int_0^1 \frac{\mathrm{e}^u}{1+u}\,\mathrm{d}u = E.\end{aligned}$$

The minus sign from $\mathrm{d}x/\mathrm{d}u$ was used to swap the limits back into increasing order.

**Ask yourself** Why should we expect the answer to be exactly $E$, with no e at all?

The map $x \mapsto \frac{1-x}{1+x}$ sends $[0,1]$ to itself (it swaps the endpoints), and the factor $\frac{1}{1+x}$ combines with $\mathrm{d}x$ to give exactly $\frac{\mathrm{d}u}{1+u}$. Indeed $\frac{\mathrm{d}x}{1+x} = -\frac{\mathrm{d}u}{1+u}$, which is a neat way to remember the whole computation.

**Part (ii)**

Now the target interval $[0,1]$ must come from $x \in [1, \sqrt{2}]$, where $x^2$ runs from 1 to 2. So we try

$$u = x^2 - 1, \qquad \mathrm{d}u = 2x\,\mathrm{d}x,$$

with $u = 0$ at $x = 1$ and $u = 1$ at $x = \sqrt{2}$. Then

$$\frac{\mathrm{d}x}{x} = \frac{x\,\mathrm{d}x}{x^2} = \frac{\mathrm{d}u}{2(1+u)}, \qquad \mathrm{e}^{x^2} = \mathrm{e}^{1+u} = \mathrm{e}\cdot\mathrm{e}^u,$$

so

$$\int_1^{\sqrt{2}} \frac{\mathrm{e}^{x^2}}{x}\,\mathrm{d}x = \frac{\mathrm{e}}{2}\int_0^1 \frac{\mathrm{e}^u}{1+u}\,\mathrm{d}u = \frac{\mathrm{e}E}{2}.$$

**Ask yourself** Why not substitute $u = x^2$?

That would work too, giving $\frac{1}{2}\int_1^2 \frac{\mathrm{e}^u}{u}\,\mathrm{d}u$, but the interval is $[1,2]$ rather than $[0,1]$. Shifting by one, $u \mapsto u-1$, lands on the same answer; $u = x^2 - 1$ simply does both steps at once.

## Checks

Simpson's rule gives $E \approx 1.125386$. Numerically, the first integral is $0.592896 = \mathrm{e} - 1 - E$, the second is $0.407104 = 2 - \mathrm{e} + E$, the integral in (i) is $1.125386 = E$, and the integral in (ii) is $1.529558 = \frac{1}{2}\mathrm{e}E$. A quick sanity check without a computer: the first two integrands are both positive, and indeed $\mathrm{e} - 1 - E \approx 0.59 > 0$ and $2 - \mathrm{e} + E \approx 0.41 > 0$; also the second is smaller than the first, as it should be since $x^2 \leq x$ on $[0, 1]$.

## Common slips

- Forgetting to reverse the limits (or dropping the minus sign) in (i), which gives $-E$, an impossible answer for a positive integrand.
- In (ii), losing the factor e from $\mathrm{e}^{x^2} = \mathrm{e}^{1+u}$, or the factor $\frac{1}{2}$ from $\mathrm{d}u = 2x\,\mathrm{d}x$.

## Beyond STEP

Integrals like $E$ are tied to the *exponential integral* $\mathrm{Ei}(x) = \int_{-\infty}^{x} \frac{\mathrm{e}^t}{t}\,\mathrm{d}t$: substituting $t = 1 + x$ shows $E = \mathrm{e}^{-1}\big(\mathrm{Ei}(2) - \mathrm{Ei}(1)\big)$. Liouville's theory shows that $\int \frac{\mathrm{e}^x}{x}\,\mathrm{d}x$ has no elementary antiderivative, which is exactly why the question treats $E$ as a named constant and asks us to reduce everything to it. Siklos, *Advanced Problems in Mathematics*, has many more questions in this spirit.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.